Lab 6

Case Study:
A Cluster on the Surgical Ward

Day 2 · 13:15–15:00 · 105 min
Groups of 4

LO3–LO5: combine genomic and epidemiological evidence to decide whether an outbreak is happening, and communicate the decision

Open your handout

Lab 6 handout

`Lab6_ssi_outbreak_case_study.pdf`

- **Download:** github.com/aryakaul/ughe-ssi-genomics-course
- Open the `labs/pdf/` folder and click the file

> The handout has every step. Copy reads, URLs and commands from the **PDF**, not from this screen.

About this case

- The **genomes are real**: MRSA isolates from a published outbreak investigation
- The **hospitals, wards, patients and dates are fictional**, so the case feels like our setting

> The real story is revealed at the end.

THE SITUATION

District Hospital A · Monday, 16 March 2026

The IPC nurse writes to you:

> *"Since early February we've had **more MRSA wound infections than usual on Surgical Ward B**. The surgeons say it's bad luck. Theatre staff say it's the ward. The ward says it's the theatre.*

The question

> *The director asks whether we should **close Ward B**. That would cancel about 40 operations a week. We got genomes back last week for all MRSA isolates since September.*
>
> ***Can you tell us what's going on by Wednesday's IPC meeting?"***

Your job

KEY MESSAGE

Answer one question for the director: should we close Ward B?

Your evidence:

1. **Pathogenwatch collection** from Lab 5: tree, timeline, ward, theatre, dates
2. **SNP matrix:** `koser_snp_matrix.tsv` (opens in Excel)

Group roles

Role	Job
Genomics lead	Reads the tree and SNP matrix
Epidemiologist	Reads the timeline: who was where, when
Clinician / IPC	How could MRSA spread? What to do?
Communicator	Writes and gives the 5-minute briefing

Plan for this lab

Part	What	Time
A	Look at the cases	15 min
B	Find the cluster	25 min
C	Genomics + epidemiology together	20 min
D	Prepare the briefing	25 min
E	Briefings to the IPC committee	15 min

Look at the cases

Open your Pathogenwatch collection: **timeline** + **data table**

QUESTIONS

1. How many MRSA isolates in total? How many from **Ward B since 1 February**?
2. Every isolate is MRSA and cefoxitin-resistant on the **lab report**. Are they all one outbreak?

Find the cluster

1. **Tree:** which isolates sit on very **short branches**?
2. **SNP matrix** in Excel: Conditional Formatting → Highlight Cells → **Less Than** `100`
3. Look for the **gap**: a group close to each other, everything else far away
4. Do the tree and the matrix agree?

PART B

Why look for a gap?

- Published thresholds for *S. aureus* are about **15–30 SNPs** (L4)
- They are **guides, not laws**

KEY MESSAGE

The clearest sign: a group close together, with a big gap to everything else.

Genomics + epidemiology together

Colour the tree by **ward**, then **theatre**. For each **Ward B patient since February**:

Patient	In the cluster?	Theatre	Linked?
P01	Yes	Theatre 2	**Linked**
P02 …			

Use the full table in your handout.

PART C

Three questions

1. **Are all the Ward B cases one outbreak?**
   Look closely at **P08**.
2. **Is anyone in the cluster not on Ward B?**
   What do they share with the others?
3. **So, should we close Ward B?**
   Where is the outbreak most likely coming from?

FINISHED EARLY?

See the extra questions in your handout.

Prepare a 5-minute briefing

For the IPC committee: director, chief surgeon, nursing lead, lab head. **Not genomics experts.**

RULES

- One flip-chart page or 3 slides
- Explain any jargon: “SNP” → “genetic differences”
- **No more than one figure**

PART D

Briefing template

1. **Bottom line:** “We found / did not find evidence of an outbreak of ___ linked to ____.”
2. **What we did:** “We sequenced __ MRSA isolates from __ to __.”
3. **What we found:** one visual + 2–3 bullets
4. **What it means:** which cases are linked; the likely route
5. **Recommendations:** 3 actions, with who and by when
6. **What we don’t know yet**

Briefings to the IPC committee

Each group has **5 minutes**. The committee must ask at least one question:

“Should we close Ward B?”

“Is it the surgeons' fault?”

“How sure are you?”

“What will this cost?”

Pause

Instructor: don’t go to the next slide until every group has given its briefing

The reveal

The genomes come from **Köser et al., *NEJM* 2012**: a suspected MRSA outbreak in a **neonatal unit** in Cambridge, UK.

- Sequencing separated the **outbreak isolates** from unrelated MRSA that looked identical on routine tests
- A follow-up study (**Harris et al. 2013**) traced transmission into the community and found a **staff carrier**, which ended the outbreak

Genomic data: ENA project PRJEB2912 (Köser et al. 2012). Scenario metadata are fictional and created for teaching.

Take-home messages

1. **Same ward + same lab result ≠ same outbreak**
2. Genomics finds **hidden links**: shared theatre, staff, equipment
3. Genomics can **prevent unnecessary actions**, like closing a ward
4. Genomics is only useful with **good metadata** and **clear communication**